

Name: _____

Solutions

Date: _____



SAFETY BAY
SENIOR HIGH SCHOOL
imagine believe achieve

Year 9 Mathematics

Test 1, 2015

Topic – Rates, Graphs, Indices



46 = _____ %
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Total Time: 60 minutes

Total Reading: 3 minutes

Total Working: 57 minutes

Weighting: 10% of the year.

Equipment:

page notes (A4 one side), Scientific Calculator

SECTION 1: CALCULATOR FREE

Time: 30 minutes	Marks for Section 1 26
Reading: 2 minutes	Equipment
Working: 28 minutes	Allowed: page notes (A4 one side)

1. [4 marks: 1, 1, 1, 1]

Simplify each of the following

a) $a^3 \times a^6$

$a^9 \checkmark$

b) $5m^4 \times 7m^5$

$35m^9 \checkmark$

c) $2xy^5 \times x^3y^2$

$2x^4y^7 \checkmark$

d) $8d^7e^3 \times d^2 \times 3de$

$24d^{10}e^4 \checkmark$

2. [4 marks: 1, 1, 1, 1]

Simplify each of the following

a) $\frac{y^6}{y^2}$

$y^4 \checkmark$

b) $\frac{21a^8}{3a^6}$

$7a^2 \checkmark$

c) $\frac{4c^{11}}{24c^4}$

$\frac{c^7}{6} \checkmark$

d) $\frac{36k^{15}}{15k^8}$

$\frac{12k^7}{5} \checkmark$

3. [5 marks: 2, 3]

Simplify each of the following

a) $\frac{4p^3 \times 8p^6}{2p^2}$

$$16p^7 \checkmark \checkmark$$

b) $\frac{3a^3 \times 4b^5 \times c^7}{5a \times b^5 \times 6c^4}$

$$\frac{12a^3b^5c^7}{30b^5c^4} \checkmark = \frac{2a^3c^3}{5} \checkmark \checkmark$$

4. [3 marks: 1, 1, 1]

Simplify each of the following

a) $(f^3)^{11}$

$$f^{33} \checkmark$$

b) $(m^5n^2)^6$

$$m^{30}n^{12} \checkmark$$

c) $(p^3)^2 \times (q^5)^4$

$$p^6q^{20} \checkmark$$

5. [4 marks: 2, 2]

Simplify each of the following

a) $\left(\frac{2d^5}{e^7}\right)^3$

$$\frac{8d^{15}}{e^{21}} \checkmark \checkmark$$

(-1 for 1 small error)

b) $\left(\frac{-2a^2b^9}{ab^7}\right)^4$

$$(-2ab^2)^4 = 16a^4b^8 \checkmark \checkmark$$

accept number in index form.

$$(2^3)$$

$$(-2)^4$$

6. [2 marks: 1, 1]

Write each of the following using a negative index

a) $\frac{1}{7}$

7^{-1}

✓

b) $\frac{1}{x}$

x^{-1}

✓

7. [4 marks: 2, 2]

Use index laws to simplify each of the following, express your answer with **positive indices**.

a) $\frac{24w^3}{16w^7}$

$\frac{3}{2w^4}$

✓✓

b) $7a^4b \times 3a^{-6}b^3$

$21a^{-2}b^4$

$= \frac{21b^4}{a^2}$

✓✓

~ END OF TEST SECTION 1 ~

Name: Solutions

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time:	30 minutes	Marks for Section 2:	20
Reading:	1 minutes	Equipment Allowed:	page notes (A4 one side), Scientific calculator
Working:	29 minutes		

Direct Proportion

$$y = kx$$

$$y = kx^n$$

Inverse Proportion

$$y = \frac{k}{x}$$

$$y = \frac{k}{x^n}$$

8. [4 marks: 1, 1, 1, 1]

State which of the following could be examples of direct proportion, inverse proportion or neither.

<i>a</i>	1	2	4	8	16
<i>b</i>	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$

Inverse ✓

<i>d</i>	1	2	5	8	10
<i>f</i>	3	12	75	192	300

Direct ✓

<i>p</i>	1	2	3	4	5
<i>z</i>	4	8	12	16	20

Direct ✓

<i>g</i>	1	2	3	4	5
<i>s</i>	4	2	1	2	4

Neither ✓

10. [4 marks: 2, 2]

Find k , the constant of proportionality if:

- a) a is directly proportional to b , and $a = 275$ when $b = 25$

$$k = \frac{275}{25}$$

$$= 11 \quad \checkmark\checkmark$$

- b) m is directly proportional to n , and $m = 42$ when $n = 28$

$$k = \frac{42}{28}$$

$$= 1.5 \quad \checkmark\checkmark$$

9. [6 marks: 2, 2, 2]

y is directly proportional to x , and $y = 24$ when $x = 6$.

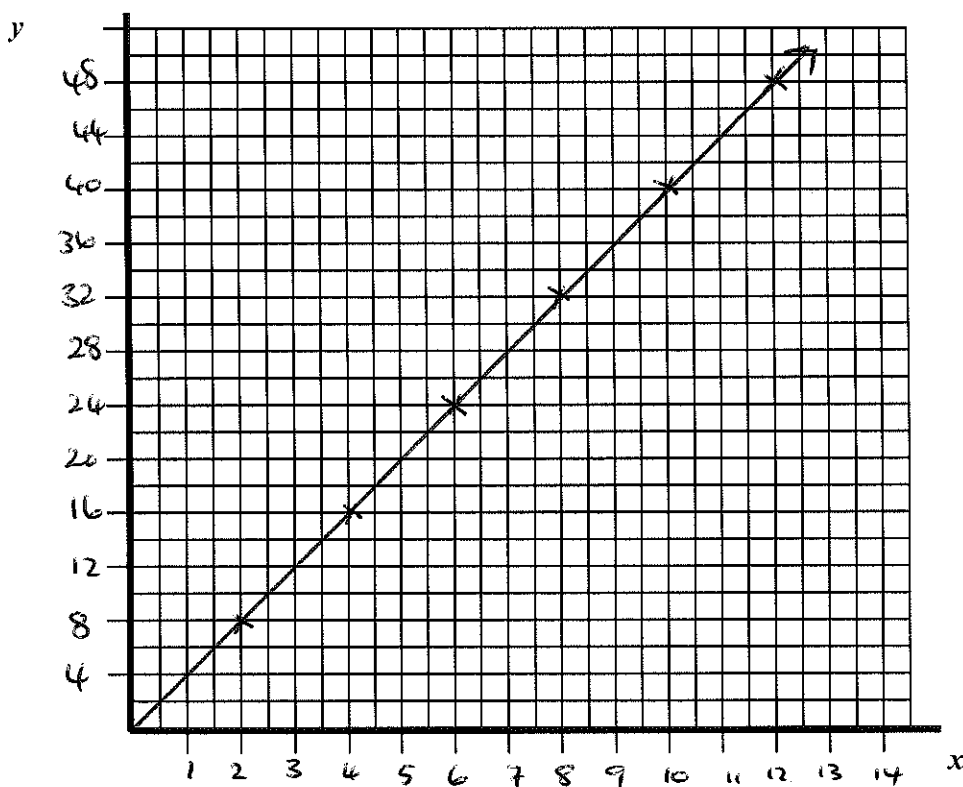
a) Find k , the constant of proportionality and the rule relating x and y

$$k = \frac{24}{6} \\ = 4 \checkmark \checkmark$$

Complete the table below by using your value of k

x	2	4	6	8	10	12
y	8	16	24	32	40	48

b) Plot the graph relating x and y



$-\frac{1}{2}$ per error.

* needs to be consistent with answer to a)

* must be a direct pattern

$\frac{1}{2}$ straight line with arrowhead going up

$\frac{1}{2}$ starts at zero

$\frac{1}{2}$ correctly plotted

$\frac{1}{2}$ consistent scale used.

11. [6 marks: 2, 2, 2]

y is inversely proportional to x , and $y = 4$ when $x = 6$.

- a) Find k , the constant of proportionality and the rule relating y and x

$$k = 4 \times 6$$

$$= 24 \checkmark$$

$$y = \frac{24}{x}$$

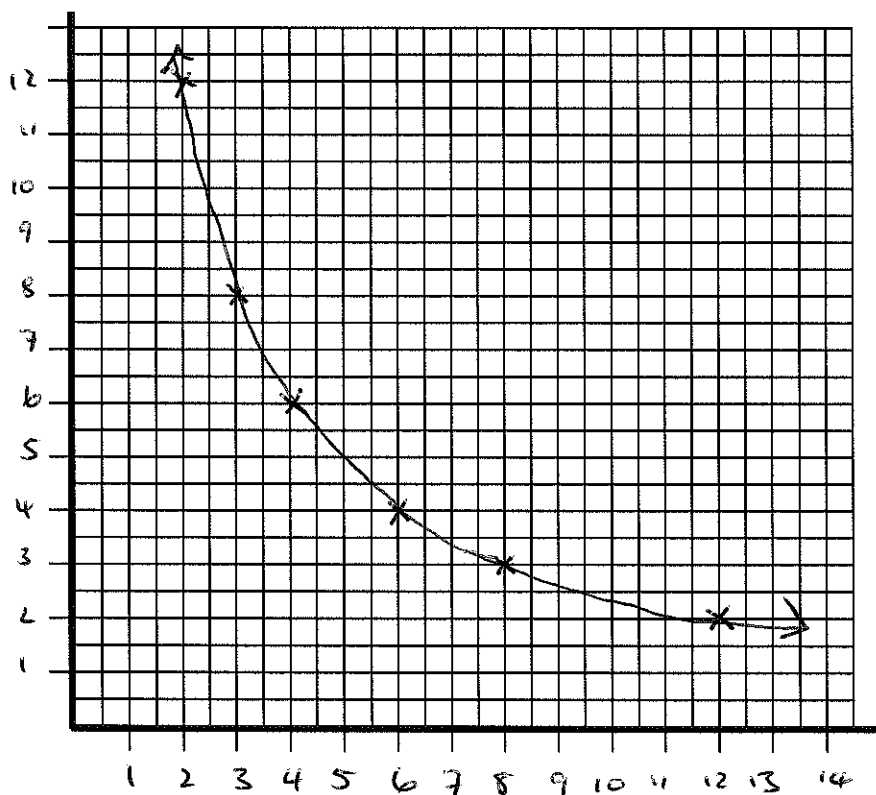
Complete the table below by using your value of k

x	2	3	4	6	8	12
y	12	8	6	4	3	2

✓✓

- b) Plot the graph relating y and x

$-\frac{1}{2}$ per error*



* needs to be considered with answer to a)

* must be an inverse pattern.

$\frac{1}{2}$ Smoothish Curve passing through points

$\frac{1}{2}$ Arrow heads used

$\frac{1}{2}$ Correctly plotted

~ END OF TEST SECTION 2 ~

$\frac{1}{2}$ Consistent scale used.